

Comparative evaluation of hemispheric and masticatory laterality in complete dentures with two occlusal schemes – A prospective cohort study

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ABSTRACT

Purpose: The lateralization can influence the choice of occlusion scheme, patient comfort and success of the prosthesis. The existence of favored masticatory side in complete denture patients and its influence with various occlusal schemes is less analyzed in the literature. The study objective was to compare the masticatory and hemispheric laterality in complete denture subjects rehabilitated with 2 different occlusal schemes at different time intervals.

Methodology: The cohort study recruited 26 patients in each group of balanced and non-balanced occlusion with definitive criteria. Conventional procedures were followed in denture fabrication. The hemispheric and masticatory laterality for all participants were established at 0,1,3- and 6-month intervals. The laterality was categorized as preferred chewing side (CPCS), predominant preferred chewing side (PPCS) and observed preferred chewing side (OPCS). The data on chewing side preference was analyzed by chi square test. ($\alpha = 0.05$).

Results: Right side preference was predominant in 86.1% of non-balanced and 60.1% of balanced occlusion participants. The side preference among the 2 laterality and across time interval decreased in masticatory laterality of balanced occlusion participants ($P < .05$) when compared to non-balanced occlusion. ($P > .05$)

Conclusion: The balanced occlusion dentures had lesser masticatory side preference when compared to non-balanced occlusion complete denture.

1. Introduction

Laterality represents a behavioral inclination towards utilization of one side of the body. It is inborn and influenced by many factors like genetics, evolutionary precedent, and environmental factors.^{1–3} The laterality is majorly controlled through cerebral cortical centers. Literatures found strongest correlation between the laterality and mastication. The preferred chewing side (PCS) or preferred mastication side (PMS) or masticatory lateral preference (MLP) is the side where jaw habitually moves during closing phase of mastication is an act of laterality. PCS are controlled by dominant cerebral hemisphere and the function can be altered by subjective perception or patient's choice.^{4–7}

The influence of PCS on various stomatognathic functions are observed in the literature. The studies on dentulous subjects displayed heavier masticatory forces in PCS.^{8,9} Takahashi et al.¹⁰ witnessed PCS in dentulous patients had wider occlusal contact area and forces are distributed over broader area. Santana-Mora et al.,¹¹ Jiang et al.,¹²

Tiwari et al.,¹³ Padmaja et al.,¹⁴ observed the differences in occlusal force due to PCS and its influence in temporomandibular joint disorders (TMD). Studies established that PCS can be influenced by peripheral factors like occlusal contact, occlusal forces, pain and TMD.^{9–11} Various studies found masticatory laterality as a major peripheral laterality test for estimating the activity of cerebral centers.^{12–14} Khamnei et al.,² Nissan et al.,¹⁵ Rovira-Lastra et al.,¹⁶ Serel Arslan et al.,¹⁷ observed relationship of PCS to hemispherical laterality test of eyeness, footedness, handedness and other laterality tests. The awareness on PCS and its testing methods can assist in establish better occlusal contacts and scheme for prosthesis.^{18–22} Additionally, it identifies the co-existence of associated ailments related to natural dentition occlusion.^{23–28}

Variability were observed in the literature on the superiority of complete denture occlusal schemes in relation to masticatory efficiency and patient comfort.^{29–33} The PCS can be a determinant factor in establishment of superiority between the occlusal schemes.⁹ The understanding on these associations can alter the science and clinical

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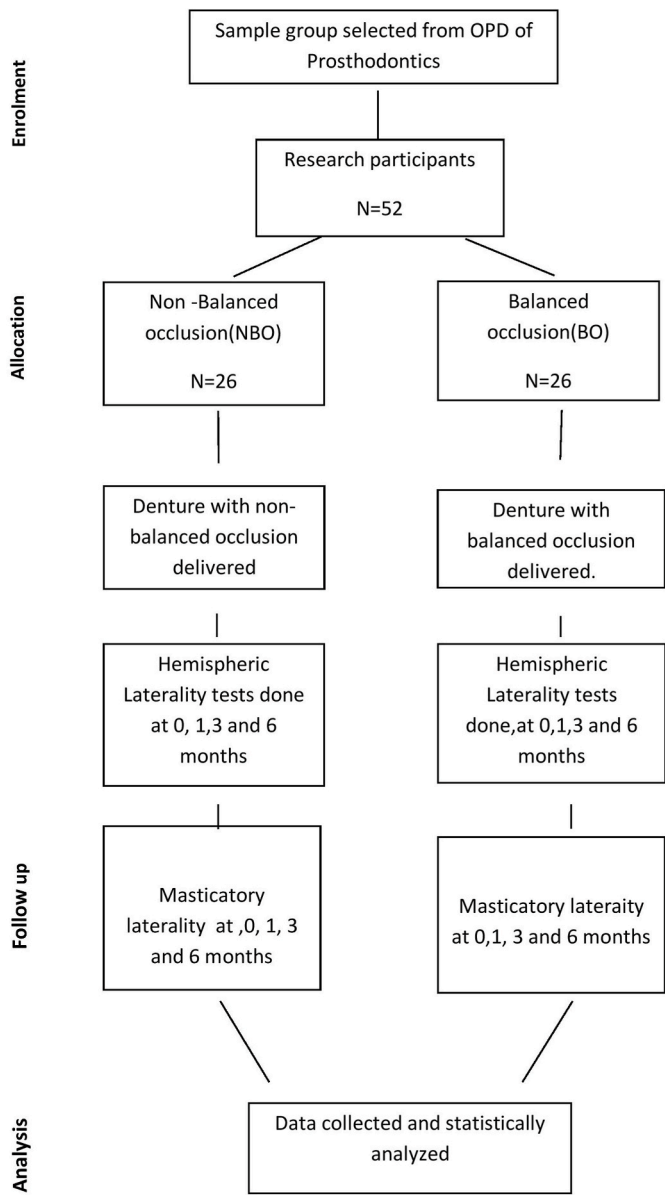


Fig. 1. Study flow chart.

perception of complete denture occlusion schemes. The establishment of these relationships can play a crucial role in deciding choice of occlusal scheme, determining deflective contacts, improving the function and patient satisfaction. These influential factors on complete denture occlusion are critical and less analyzed in literature. The objective of the study was to compare the laterality at various time intervals on complete denture participants, rehabilitated with non-balanced and balanced occlusion. The null hypothesis of the study was there is no difference in the laterality of complete dentures rehabilitated with non-balanced and balanced occlusal schemes.

2. Materials and methods

The study was approved by Institutional ethical committee (SRMDC/IRB/2019/MDS/No.206). The study was conducted in department of prosthodontics, SRM Dental College, Ramapuram between January 2020 to March 2021. The study was designed with 80% power, 5% alpha error and estimated sample size from the literature was 26 in each group (n = 52).²¹ The study followed convenient sampling technique. The participants consent was obtained and were allotted to the test groups by

Table 1
Tests for laterality.

Organ	Custom objects	Observation
Ear	Hollow cone	Side of ear used to listen to sound generated through hollow cone
	Mobile phone	Side of ear used to speak over phone
Eye	Opaque bottle	side of eye used to look through opaque bottle
	Dark box	side of eye used to look through the dark box
Mastication	Chewing gum	side to which the gum is spotted after 15 s chewing.

odd/even numbering system (Fig. 1). The participants were selected with definitive inclusion and exclusion criteria. Edentulous patients of both sexes between 45 and 75 years, class I category of Prosthodontic Diagnostic index (PDI), no congenital or acquired defects in hearing and eyesight except refractory errors were considered.³⁴ Patients with jaw dysfunction, severe facial asymmetry, TMJ disorder, malignant diseases, active infection, allergic to the materials, uncooperative patients, autoimmune diseases, congenital or acquired defects in hearing or in eyesight except for refractive errors, were excluded from the study.

The denture was fabricated by the primary investigator following the conventional procedures. The non-balanced occlusion dentures were fabricated on 3-point articulator and semi-adjustable articulator (Hanau 010885000 Wide-Vue, Whip mix, USA) was used to fabricated balanced occlusion dentures. Standardized clinical and laboratory procedures were followed from pre to post denture insertion.³⁵ The patients were blinded over the entire procedure. The side preference of peripheral organ function of eye and ear were recorded using hemispheric laterality tests.^{16–21} The study used customized objects and employed two dissimilar commands for each organ to determine laterality (Table 1). A hollow cone, mobile phone, opaque bottle and a dark box with a hole on one side were placed on a table and selected at free will by the patients. The procedures were explained to the participants and were allowed to select the objects at will. The involuntary right or left side preference of organ for the test object was documented. The Asymmetrical (I) index was estimated from the data obtained. I index was calculated with the equation

I= (R-L)/(R + L)'

R - The number of observations made on right side
L- The number of observations made by the left side

Laterality interpretation varied with the numerical results of I index. The results ranged from +1 to –1. The positive result between +0.1 and +1 favored the right side. The negative results ranging from –0.1 to –1 expressed left side dominance.

MLP was determined by visual spot-checking method. Participant were given a gum to chew for 15 s and were asked to smile to observe the position of bolus.^{4,14,24,36,37} The method was repeated seven times and preferred side of holding the bolus was recorded.¹⁵ The masticatory laterality was classified as consistent preferred chewing side (CPCS) if the bolus was found on the same side for 7/7 of the chewing cycles. Preferred chewing side (PPCS) was perceived if bolus was found on the same side for 5/7 or 6/7 chewing cycles and observed preferred chewing side (OPCS) was resolved based on maximum chewing side preference. The estimation of hemispheric and masticatory laterality for each occlusal scheme group was compared for four-time intervals 0, 1, 3 and 6month after denture placement (Fig. 1). All observations were made thrice by 3 observers who were not involved in the denture fabrication and the mean of the observations was recorded for all the tests and at various time intervals. The relationship between/within the group at each interval was determined. The raw data was tabled and statistically

Table 2
Demographic distribution of data.

Group	N	Male	Female	Mean Age	Std. Deviation	Std. Error Mean	P value
Non-balanced group	26	11	15	64.38	8.39	1.64	0.424
Balanced occlusion group	26	8	18	62.61	7.39	1.45	

analyzed with Chi square test.

3. Results

The demographic distribution of participants of age and gender is listed in Table 2. The mean age of the participants ranged from 62.61 ± 7.39 years in non-balanced group and 64.38 ± 8.39 years in balanced occlusion test group. The test showed that there was no significant difference in the age, gender distribution between the two groups. ($P > .05$).

The laterality of non-balanced occlusion at each time interval and summary for overall time period established predominant right-side laterality (Table 3). The mean masticatory laterality in balanced occlusal group, at the time of insertion was 92.3% right side and 7.7% displayed left side preference. The laterality in balanced occlusion group displayed increased no preference side in the masticatory laterality over the time intervals. At 6th month interval, 96.2% of the balanced occlusion patients showed no side preference during mastication, 3.8% displayed a right-side preference and no left side preference was observed. The combined laterality at 6th month displayed 48.1% no side preference in masticatory laterality of balanced occlusion group. (Table 4).

The statistical analysis of both the intervention groups at various time intervals were statistically significant. ($P < .05$) Chi-square test analysis (Table 5) of laterality for various time intervals was statistically

significant for balanced occlusion for masticatory laterality ($P < .05$) and it was insignificant for other tests of laterality. ($P > .05$) The association of hemispheric and masticatory laterality with-in the groups (Table 5) was statistically insignificant during denture insertion for both denture groups and at various time intervals for non-balanced denture group. ($P > .05$) The association between both the laterality (Table 6) was statistically significant for balanced denture group at various time intervals. ($P < .05$).

4. Discussion

The null hypothesis was rejected due to the differences in laterality

Table 5
Assessment of laterality across the timeline.

		Timeline				P Value
		0 Month	1 Month	3 Months	6 Months	
Masticatory – Balanced	Right	24	10	3	1	.001
	Left	2	1	0	0	
	No preferred side	0	15	23	25	
Masticatory – Non-Balanced	Right	21	23	22	22	.89
	Left	5	3	4	4	
	No preferred side	0	0	0	0	
Hemispheric Laterality- Balanced	Right	21	22	22	22	.97
	Left	5	4	4	4	
	No preferred side	0	0	0	0	
Hemispheric Laterality- Non-Balanced	Right	23	23	22	23	.96
	Left	3	3	4	3	
	No preferred side	0	0	0	0	

Table 3
Percentage distribution of tests at different time intervals for non-balanced occlusion scheme.

Type of test	Side	Day of Insertion		1 Month		3 Months		6 Months		Over all	
		N	%	N	%	N	%	N	%	N	%
Brain Laterality	Right	23	88.5	23	88.5	22	84.6	23	88.5	91	87.5
	Left	3	11.5	3	11.5	4	15.4	3	11.5	13	12.5
	No preferred side	0	0	0	0	0	0	0	0	0	0
Masticatory Laterality	Right	21	80.8	23	88.5	22	84.6	22	84.6	88	84.6
	Left	5	19.2	3	11.5	4	15.4	4	15.4	16	15.4
	No preferred side	0	0	0	0	0	0	0	0	0	0
Both	Right	43	84.3	46	88.5	44	84.6	45	86.5	179	86.1
	Left	8	15.7	6	11.5	8	15.4	7	13.5	29	13.9
	No preferred side	0	0	0	0	0	0	0	0	0	0

Table 4
Percentage distribution of tests at different time intervals of balanced occlusion scheme.

Type of Occlusion	Side	Day of Insertion		1 Month		3 Months		6 Months		Over all	
		N	%	N	%	N	%	N	%	N	%
Brain Laterality	Right	21	80.8	22	84.6	22	84.6	22	84.6	87	83.7
	Left	5	19.2	4	15.4	4	15.4	4	15.4	17	16.3
	No preferred side	0	0	0	0	0	0	0	0	0	0
Masticatory Laterality	Right	24	92.3	10	38.5	3	11.5	1	3.8	38	36.5
	Left	2	7.7	1	3.8	0	0	0	0	3	2.9
	No preferred side	0	0	15	57.7	23	88.5	25	96.2	63	60.6
Both	Right	45	84.7	32	61.5	25	48.1	23	44.2	125	60.1
	Left	7	15.3	5	9.6	4	7.7	4	7.7	20	9.6
	No preferred side			15	28.8	23	44.2	25	48.1	63	30.3

Table 6

Association between hemispheric and masticatory laterality.

Type of Occlusion	Timeline	Preferred Side	Description		P Value
			hemispheric Laterality	Masticatory Laterality	
Balanced Occlusion	0 Month	Right	21	24	.22
		Left	5	2	
		No preferred side	0	0	
	1 Month	Right	22	10	.001
		Left	4	1	
		No preferred side	0	15	
	3 Months	Right	22	3	.001
		Left	4	0	
		No preferred side	0	23	
	6 Months	Right	22	1	.001
		Left	4	0	
		No preferred side	0	25	
Non-Balanced Occlusion	0 Month	Right	23	21	.44
		Left	3	5	
		No preferred side	0	0	
	1 Month	Right	23	23	1.0
		Left	3	3	
		No preferred side	0	0	
	3 Months	Right	22	22	1.0
		Left	4	4	
		No preferred side	0	0	
	6 Months	Right	23	22	0.68
		Left	3	4	
		No preferred side	0	0	

between the 2 occlusal groups at various time intervals. The results of the study established no significant variation in PCS and hemisphere laterality in non-balanced occlusion at various time intervals. ($P > .05$) The study found demonstrated that the PCS had no lateral preferences in non-balanced group. The study outcomes were similar to studies of Pihlaja et al.,¹⁸ Flores-Orozco EI et al.,¹⁹ Lee et al.²⁰

Kazazoglu et al.³⁷ found a strong favor for a particular side in mastication due to unilateral chewing, denture movements and poor alveolar ridges. It was inferred that the repositioning of the bolus improved the masticatory efficiency. The non-balanced group (Table 5) displayed higher right sided preference of 86.1%. Christensen et al.,⁴ Arslan et al.,¹⁷ Barcellos et al.,²¹ Diernberger et al.,²³ believed that incidence of higher frequency of right sidedness, irrespective of peripheral factors, indicate the influence of central nervous system mechanism. The studies established the presence of a consistent and predominant lateral preference were attributed to determinants of a motivational chewing behavior controlled by the sensorimotor systems.

The analysis in balanced occlusion group inferred an insignificant change in distribution of lateralization of organs between hemisphere and masticatory choice over the investigation period. ($P > .05$) During the initial study period a statistically significant correlation existed between the hemispheric and masticatory lateral preference in bilaterally balanced occlusion group. At six-month, there was a decline in this correlation and data were suggestive of no side preference (Tables 5 and 6). The summary analysis exhibited 30.3% no side preference and 60.1% right side preference. The results inferred no PCS in bilateral balance

occlusion dentures and it implies that the occlusal scheme incorporated made it easier for the patients to masticate without PCS. Bhambhani et al.,²⁴ Neto et al.,²⁵ Lemos et al.,²⁹ Moradpoor et al.,³¹ Goldstein et al.,³² found enhanced bilateral chewing, masticatory efficiency, psychological comfort in balanced dentures. The results of the study were supportive to the findings of the supportive studies of balanced occlusion.

Several studies evidenced that the visual observations and the association between preferred chewing side.^{6,14,30,32} Christensen et al.,⁴ Pond et al.²⁶ demonstrated improved muscle activity in chewing side when compared to non-chewing side. The improved activity validated the PCS. The literature demonstrated predominant right-side PCS than left side.^{2,3,12} Jiang et al.⁵ and DeYoe et al.²⁷ found changes in the cortices to PCS in fMRI and BOLD procedures further strengthening the association between a hemispheric dominance and chewing side preference.

Masticatory laterality improved the clinician attention towards the patient comfort, improved patient quality of life and to obtain the psychological advantage of the treatment. The hemisphere centers and neural pathways play a significant role in regulating masticatory function. It can be affected by the loss of teeth and it can indirectly affect the other activities. Hemispheric and masticatory lateral preference help to ascertain prevalence of cognition impairment. Reports exist on maintaining PCS leading to steeper condylar path angle, flatter anterior guidance angle and diseases of temporomandibular joint. Studies described chewing side preference had an effect on memory impairment and decreased brain function.¹⁶ The assessment and understanding of hemisphere and masticatory lateral preference was not only beneficial in assessing the state of patients oral and cognitive functions, but aids in treatment plan that could assist in avoiding future complications.^{16,20}

The limitations of the study where it recruited majorly right-side dominance individuals, used qualitative test methods, shorter assessment period, and compared two occlusal schemes among other occlusion schemes of complete dentures. The results of the study have to be interpreted within these study limitations.

The study has further scope for research on hemispheric and masticatory lateral preference on existing complete denture wearers. The assessment on various complete denture occlusal schemes shall be productive. The participants selected for the study could be included from a broader group by involving other categories from the ACP classification. The analysis in these parameters shall improve the understanding and it can enhance the prosthodontic implication.

5. Conclusion

The balanced occlusion scheme denture had lesser side preference when compare to non-balanced occlusion complete denture.

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